

Tier 2 Student Guide

The Power Lab as a Bridge to a Test Envelope

This example connects the familiar power lab measurement to the Tier 2 question: how does the uncertainty change when the test condition changes? Here the test condition is the resistance. The resistance can be changed by replacing the load resistor while the electrical measurement system remains the same.

1 Start with three valid power data-reduction equations

For a resistor with voltage V , current I , and resistance R , the same nominal power can be written three ways:

$$P_1 = VI, \quad P_2 = I^2 R, \quad P_3 = \frac{V^2}{R}.$$

These are physically equivalent when $V = IR$, but they do not have the same uncertainty propagation because each equation uses a different pair of measured quantities.

For any data-reduction equation $y = f(x_1, x_2, \dots)$, define the signed dimensionless sensitivity coefficient by

$$S_{x_i} = \frac{x_i}{y} \frac{\partial y}{\partial x_i}.$$

Apply that definition to each power equation. For $P_1 = VI$,

$$\frac{\partial P_1}{\partial V} = I, \quad \frac{\partial P_1}{\partial I} = V, \quad \frac{\partial P_1}{\partial R} = 0,$$

so

$$S_V = \frac{V}{P_1} I = 1, \quad S_I = \frac{I}{P_1} V = 1, \quad S_R = \frac{R}{P_1} (0) = 0.$$

For $P_2 = I^2 R$,

$$\frac{\partial P_2}{\partial V} = 0, \quad \frac{\partial P_2}{\partial I} = 2IR, \quad \frac{\partial P_2}{\partial R} = I^2,$$

so

$$S_V = 0, \quad S_I = \frac{I}{P_2} (2IR) = 2, \quad S_R = \frac{R}{P_2} I^2 = 1.$$

For $P_3 = V^2/R$,

$$\frac{\partial P_3}{\partial V} = \frac{2V}{R}, \quad \frac{\partial P_3}{\partial I} = 0, \quad \frac{\partial P_3}{\partial R} = -\frac{V^2}{R^2},$$

so

$$S_V = \frac{V}{P_3} \frac{2V}{R} = 2, \quad S_I = 0, \quad S_R = \frac{R}{P_3} \left(-\frac{V^2}{R^2} \right) = -1.$$

data-reduction equation	S_V	S_I	S_R
$P_1 = VI$	1	1	0
$P_2 = I^2 R$	0	2	1
$P_3 = V^2/R$	2	0	-1

For example, $S_R = -1$ in $P_3 = V^2/R$ means that a fractional increase in R produces an approximately equal fractional decrease in P_3 , holding V fixed. The negative sign disappears when that contribution is squared in an uncertainty budget.

For this simplified bridge example, use unequal manufacturer half-widths for the proportional part of each measurement uncertainty:

$$h_V = 0.005\% = 5.0 \times 10^{-5}, \quad h_I = 0.06\% = 6.0 \times 10^{-4}, \quad h_R = 0.012\% = 1.2 \times 10^{-4}.$$

Interpret each value as a bounded manufacturer statement of $\pm h_x|x|$. Each half-width is converted to its own standard uncertainty by dividing by $\sqrt{3}$, and those standard uncertainties are then combined by root-sum-square:

$$u_x = \frac{h_x|x|}{\sqrt{3}}.$$

The percentage values are not assumed to be equal. The fixed sensor contributions are introduced after the resistance is selected as the changing test condition, because their effect depends on the magnitude of the measured signal.

The numerical values in this bridge are simplified illustrative placeholders. They are not direct mappings from the previous power-lab activity and are not specifications for the Tier 2 instruments. Their purpose is to make the changing-test-condition idea visible with a manageable example.

2 Make resistance the changing test condition

Choose a voltage level V_0 that is safe for the circuit and use a set of resistors spanning the intended test range. For an illustrative constant-voltage study,

$$V(R) = V_0, \quad I(R) = \frac{V_0}{R}, \quad P(R) = \frac{V_0^2}{R}.$$

The actual laboratory may have a different source behavior. Use the measured or instructor-provided $V(R)$ and calculate the nominal current and power from the selected model. The important point is that R is the independent test condition, while V , I , and P generally change with R .

3 Assign systematic standard uncertainties to the instruments

Use the supplied specifications to construct an uncertainty model for each measured variable. In addition to the percentage-of-reading manufacturer half-width, include a fixed sensor contribution. This fixed term approximates reading-independent effects such as resolution, digitization, quantization, or a sensor noise floor.

$$\begin{aligned} b_{V,\text{sys}}(R) &= \frac{1}{\sqrt{3}} \sqrt{(h_V|V(R)|)^2 + d_V^2}, & h_V &= 0.005\%, \\ b_{I,\text{sys}}(R) &= \frac{1}{\sqrt{3}} \sqrt{(h_I|I(R)|)^2 + d_I^2}, & h_I &= 0.06\%, \\ b_{R,\text{sys}}(R) &= \frac{1}{\sqrt{3}} \sqrt{(h_R|R|)^2 + d_R^2}, & h_R &= 0.012\%. \end{aligned}$$

For illustration, use $d_V = 0.0025$ V, $d_I = 0.0005$ A, and $d_R = 0.10$ Ω . These are placeholder fixed half-widths, not specifications for the actual laboratory instruments. Each rectangular half-width is divided by $\sqrt{3}$ to obtain its own standard uncertainty before the terms are combined by root-sum-square; the

combined quantity is not itself a single rectangular half-width. No additional generic accuracy coefficient is being added.

The normalized systematic uncertainties of the inputs are

$$n_V(R) = \frac{b_{V,\text{sys}}(R)}{|V(R)|}, \quad n_I(R) = \frac{b_{I,\text{sys}}(R)}{|I(R)|}, \quad n_R(R) = \frac{b_{R,\text{sys}}(R)}{|R|}.$$

The percentage terms approach a constant floor when the measured signal is large. The fixed terms become increasingly important when the signal is small. For this constant-voltage sweep, current decreases as R increases, so n_I rises with R ; resistance becomes easier to measure fractionally as R increases, so n_R falls with R . Those competing trends can create a bucket-shaped normalized uncertainty curve.

4 Produce the normalized and scaled power curves

For each candidate data-reduction equation, multiply the input normalized uncertainty by the magnitude of its sensitivity coefficient:

$$\begin{aligned} n_{V \rightarrow P_1} &= n_V, & n_{I \rightarrow P_1} &= n_I, \\ n_{I \rightarrow P_2} &= 2n_I, & n_{R \rightarrow P_2} &= n_R, \\ n_{V \rightarrow P_3} &= 2n_V, & n_{R \rightarrow P_3} &= n_R. \end{aligned}$$

Combine the active contributions by root-sum-square:

$$\begin{aligned} \text{NSU}_{P_1,\text{sys}}(R) &= \sqrt{n_V^2 + n_I^2}, \\ \text{NSU}_{P_2,\text{sys}}(R) &= \sqrt{(2n_I)^2 + n_R^2}, \\ \text{NSU}_{P_3,\text{sys}}(R) &= \sqrt{(2n_V)^2 + n_R^2}. \end{aligned}$$

These are the **normalized systematic standard uncertainty** curves. They answer: what fraction of the reported power is uncertain at each resistance?

Scale each normalized curve by the nominal power:

$$\text{SSU}_{P_k,\text{sys}}(R) = |P_k(R)| \text{NSU}_{P_k,\text{sys}}(R), \quad k \in \{1, 2, 3\}.$$

These are the **scaled systematic standard uncertainty** curves. They answer: how many watts of uncertainty should be attached to the reported power at each resistance?

5 Monte Carlo version of the same power study

The Monte Carlo analysis uses the same uncertainty model as the Taylor-series analysis, but it propagates simulated measurements through the power equations directly. At each resistance, generate a large population of possible measured values. For each input, include both a percentage-of-reading perturbation and an independent fixed-term perturbation:

$$\begin{aligned} V^* &= V + h_V|V|\xi_{V,1} + d_V\xi_{V,2}, \\ I^* &= I + h_I|I|\xi_{I,1} + d_I\xi_{I,2}, \\ R^* &= R + h_R|R|\xi_{R,1} + d_R\xi_{R,2}, \end{aligned}$$

where every ξ is an independent draw from a uniform distribution on $[-1, 1]$. The two perturbations for each input are independent because they represent different uncertainty sources. The corresponding simulated powers are

$$P_1^* = V^* I^*, \quad P_2^* = (I^*)^2 R^*, \quad P_3^* = \frac{(V^*)^2}{R^*}.$$

At each resistance, calculate the sample standard deviations:

$$\text{SSU}_{P_k, \text{sys}}^{\text{MC}}(R) = \text{std}(P_k^*), \quad \text{NSU}_{P_k, \text{sys}}^{\text{MC}}(R) = \frac{\text{SSU}_{P_k, \text{sys}}^{\text{MC}}(R)}{|P_k(R)|}.$$

Use this workflow across the full resistance sweep:

1. Set the nominal $V(R)$, $I(R)$, and R .
2. Draw independent bounded uniform values for each proportional and fixed contribution.
3. Construct V^* , I^* , and R^* for every Monte Carlo trial.
4. Evaluate P_1^* , P_2^* , and P_3^* .
5. Compute SSU and NSU from the simulated populations.
6. Plot the Monte Carlo estimates over the Taylor-series curves.

The accompanying MATLAB script performs this resistance sweep and produces the plotted Monte Carlo comparisons. The Monte Carlo points should follow the Taylor-series curves when the assumptions are consistent. Small differences are expected because the Monte Carlo results are estimated from a finite number of trials. Larger differences in standard deviation would point instead to nonlinearity relative to the uncertainty size, a nominal value near zero, omitted correlations, or nonsmooth behavior. The bounded distributions mainly change output shape and percentile or coverage intervals; they do not by themselves explain a standard-deviation mismatch for a linear model.

Use independent draws only when the physical uncertainty sources are independent. If the same calibration error affects multiple readings, preserve that correlation in the Monte Carlo model. The bounded uniform sampling here is a deliberate simplification that matches the manufacturer half-width assumption; it is not a replacement for the actual uncertainty distributions in the Tier 2 laboratory.

6 Optional MATLAB reproduction

The assignment already provides the three illustrative plots below; you are not required to generate them for your report. The complete, presentation-quality script is available here: `open matlab_code.m`. You may also download the raw MATLAB file and run or modify it locally. Replace the illustrative instrument values only if you want to explore a different sensor model.

7 Pre-generated MATLAB plots

The following plots use the illustrative manufacturer half-widths and fixed sensor contributions in the accompanying MATLAB example. They are provided so you can focus on interpreting the uncertainty curves and the resulting test-envelope recommendation. The values are examples, not specifications for the actual laboratory instruments.

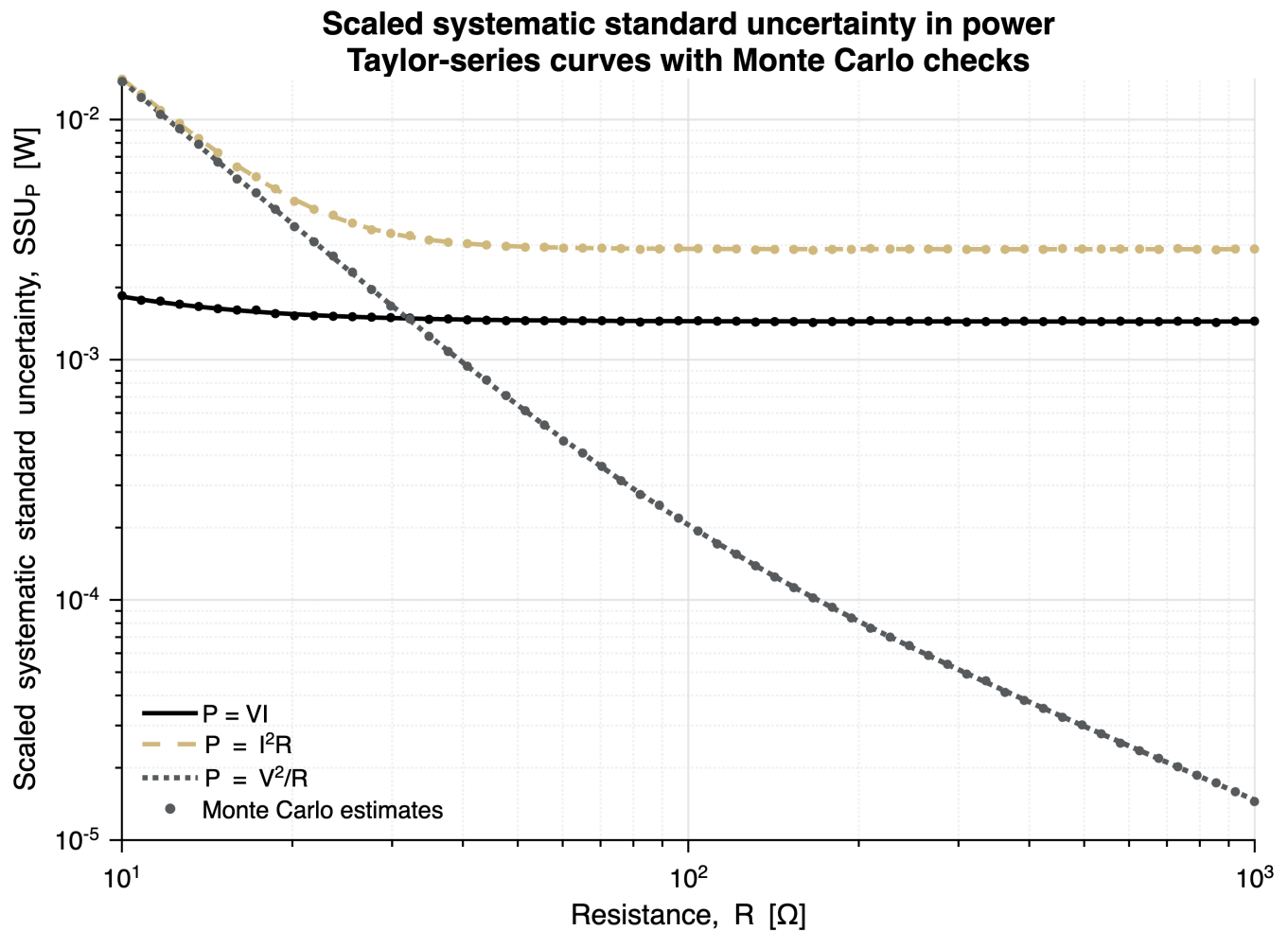


Figure 1: Scaled systematic standard uncertainty in power versus resistance.

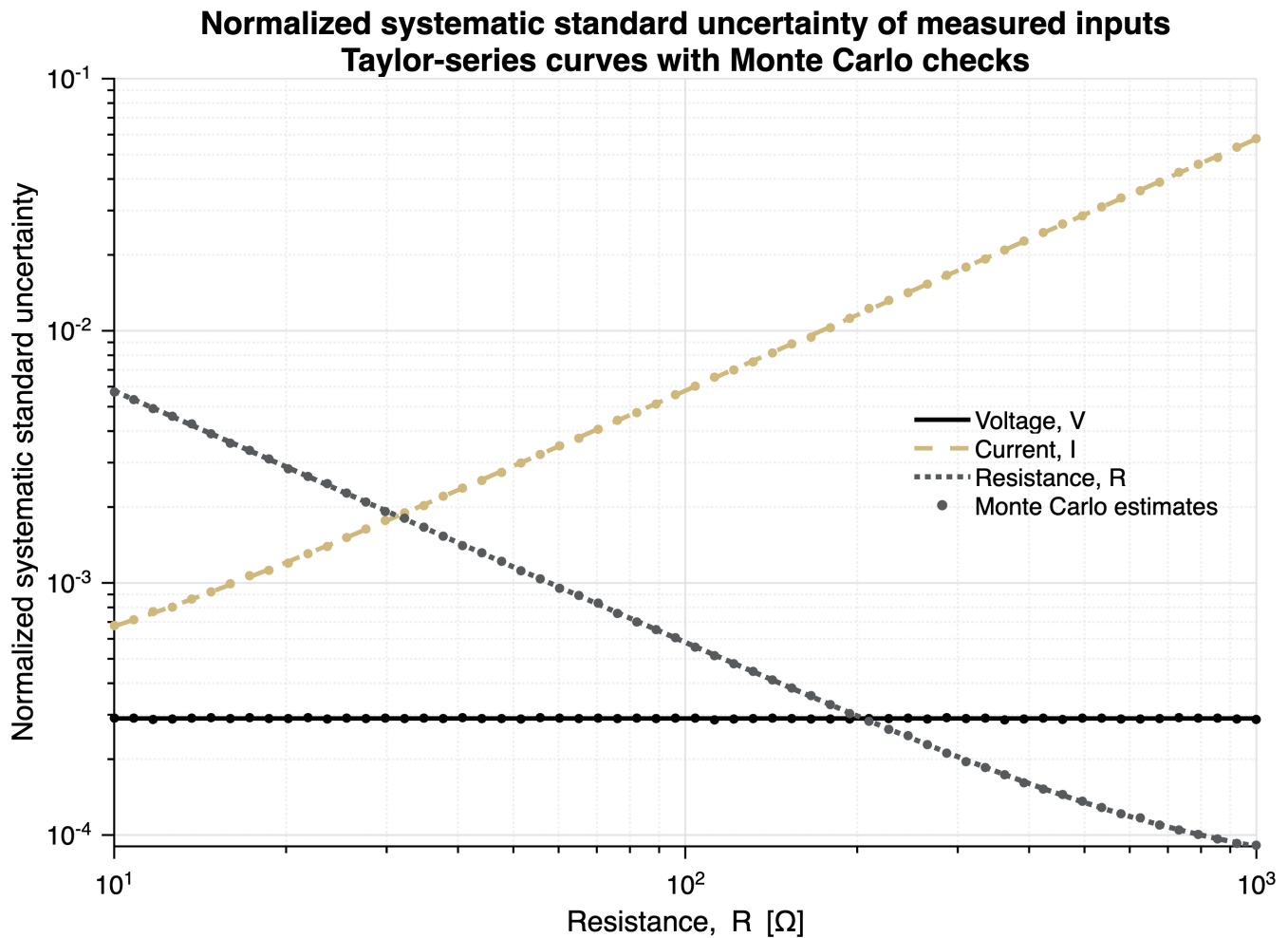


Figure 2: Normalized systematic standard uncertainty of voltage, current, and resistance versus resistance.

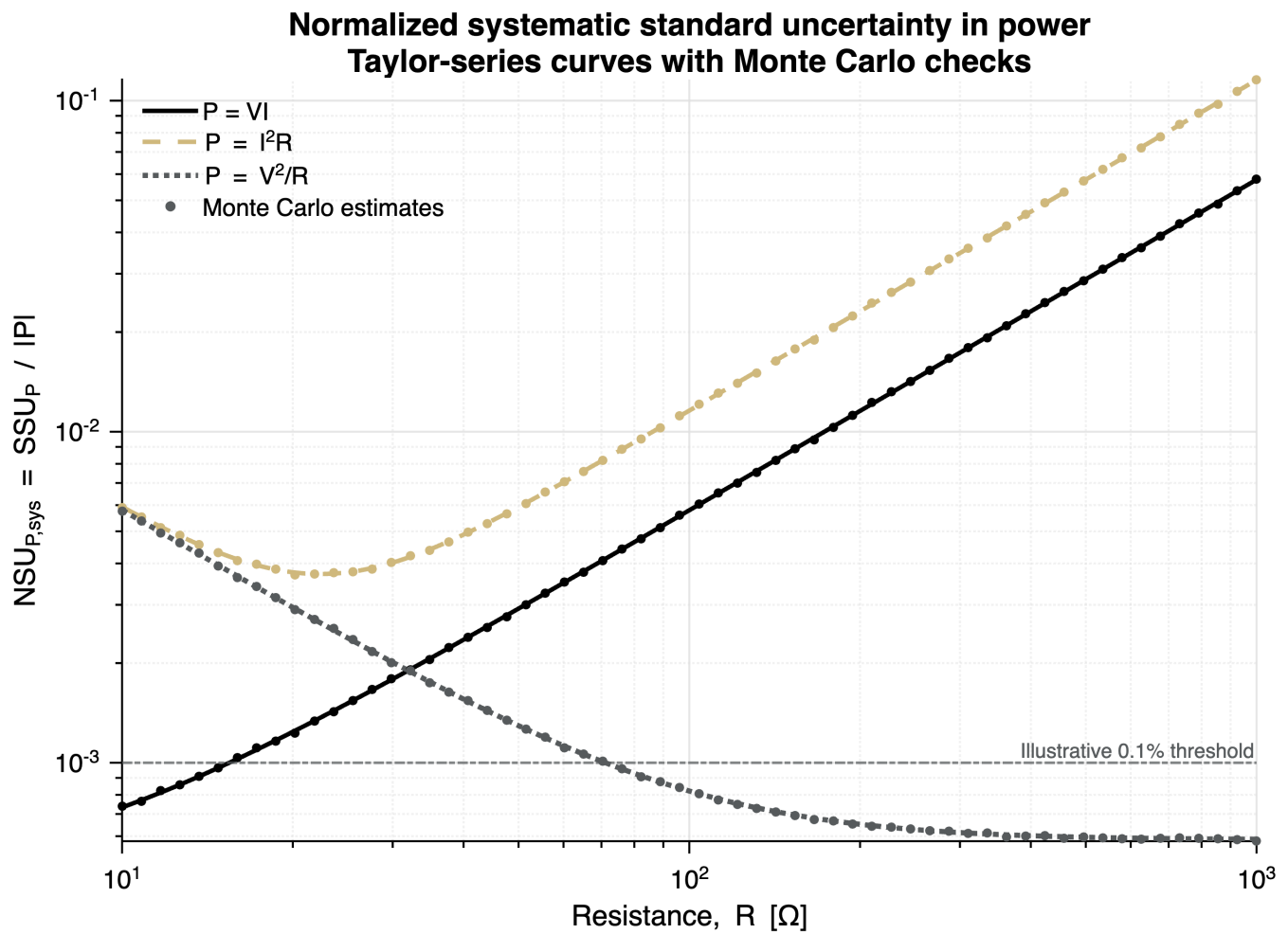


Figure 3: Normalized systematic standard uncertainty in power versus resistance, including Monte Carlo checks.

8 What the plots mean

The fixed sensor contribution creates the curvature. At low signal levels, resolution and digitization can dominate the percentage uncertainty; at larger signal levels, the percentage-of-reading term becomes the limiting floor. In the $P_2 = I^2 R$ expression, the current contribution grows with resistance while the resistance contribution decreases, producing a bucket-shaped region of lower normalized uncertainty.

The main lesson is that the data-reduction equation itself moves the useful test envelope. In this illustrative case, $P = VI$ is preferred at lower resistance, while $P = V^2/R$ becomes preferred at higher resistance; $P = I^2 R$ is not the best choice everywhere. The crossover depends on the actual sensor specifications and operating constraints. Here, V^2/R wins mainly because the placeholder voltage coefficient h_V is the smallest. The best test region is where the scaled power uncertainty is acceptably small, the normalized uncertainty is below the client threshold, the electrical limits are satisfied, and the selected data-reduction equation is appropriate.

Do not choose a resistor only because it minimizes relative uncertainty. A very high resistance may produce too little current for the ammeter, while a very low resistance may exceed the source or resistor power rating.

9 Foundational references

The power example is a concrete application of the broader measurement-uncertainty framework summarized in the companion guide. For students who want to go deeper:

- **Coleman and Steele, *Experimentation, Validation, and Uncertainty Analysis for Engineers*, 4th ed.** Wiley, 2018. Chapters 1, 3, and 4 support the uncertainty-source, Taylor-series, Monte Carlo, and experiment-planning ideas used here. [Wiley Online Library](#).
- **JCGM 100:2008, *Evaluation of measurement data - Guide to the expression of uncertainty in measurement* (GUM).** The international reference for evaluating and expressing measurement uncertainty. [BIPM publication page](#).
- **Taylor and Kuyatt, *Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results*, NIST Technical Note 1297.** A practical reference for standard uncertainty, Type A and Type B evaluations, and combined uncertainty. [NIST Technical Note 1297](#).
- **Possolo, *Simple Guide for Evaluating and Expressing the Uncertainty of NIST Measurement Results*, NIST Technical Note 1900.** A more recent NIST guide with worked examples and additional Monte Carlo discussion. [NIST Technical Note 1900](#).
- **ASME PTC 19.1, *Test Uncertainty*.** A professional test-engineering standard for propagating uncertainty into test results. [ASME standard page](#).

These are background references rather than additional required readings. The companion normalized-uncertainty guide explains how the general framework is being simplified for this Tier 2 example.

10 The Tier 2 translation

Power example	Aeroelasticity	Spin Modules
Test condition R	Freestream velocity or angle of attack	Applied torque or angular rate
Measured inputs V, I, R	Differential pressure, atmospheric pressure, temperature, angle of attack	Current, gyro rate, encoder rate, motor torque constant
Derived output P	Airspeed, C_l , or C_d	Predicted base rotational rate and encoder residual
Low-signal problem	Small differential pressure at low speed	Small net torque or low gyro rate
Improvement	Pressure-sensor range, resolution, calibration, or sampling	Current/torque range, encoder measurement, timing, or resistance characterization

The analysis is not complete until the group identifies the dominant input, marks the useful and inappropriate regions, and proposes a specific improvement that moves the test envelope.